

Lecture 45

Cyber Security and Privacy - Capstone Discussion A

Cyber Security and Privacy | NPTEL 106106248

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Video: <https://www.youtube.com/watch?v=HXt6GvYxSqQ>

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you can start sir hello hello am I audible good morning can hear yeah yes sir okay okay I can have some technical snack but I hope everything is fine now so good morning and and welcome to this live session I want to believe all of you are enjoying the lessons enjoying the chapters that you are learning week on week and you find the assignments interesting and we've been receiving some feedback and we've been responding to your questions on the chat box and we of course look forward to receiving more questions and more feedback and we are happy to help you wherever we can and some things as I said before if it's related to your career aspirations Etc we may you may find our answers sometimes not very useful but nevertheless career is something that you discover in a journey so do stay on and post us more questions so that you we also remain engaged throughout the week and we also want we want you also to be engaged and engage us too and so that this becomes a very learning full journey and I guess you are into the eth week so we were just reviewing the lessons that you have already learned so you are going soon going to switch to information privacy as a as another module in this course but you have almost completed all the lessons that are relevant to cyber security management and cyber security policy more from A administrative or techno managerial point of view and you have already understood that this is not a technical course but technology is one resource that is involved in this in in in this space called cyber security but that's not the only thing we also have seen there are other stakeholders and other aspects that are involved in cyber security management which involves people which involves policies and processes standards and Frameworks so that is what we have seen so far and my effort in this course is to give a decent foundation on cyber security and particularly Concepts okay for example in the slide that is shown here you're seeing some key words like threats attack vulnerability and risk okay and often times these words are used interchangeably when people talk about cyber security they may use these terms as if they mean the same okay so you have seen now having gone through the es You Now understand what is a threat threat is not a risk okay threat is a source of risk okay risk is the ultimate U thing that happen but it is actually threats that lead to a state of risk okay risk is like a state okay attack is an event that occurs at a point in time okay and attack happens because there is a threat and threat has a certain probability okay so there are also chances involved okay just because it's U going to rain it doesn't mean that there going to be a flood okay even if there is heavy rain if you are prepared then you know that there cannot be a flood or the the water is not going to really cause any major risk to you okay so there is also something called preparedness which actually brings us to the topic of residual risk okay we are actually prepared but despite our well preparedness there could be major events which can actually sort of go beyond our capability to protect ourselves okay so that is called residual risk or what that which is left over despite our current

preparation and also there is vulnerability okay so vulnerability is okay so you have threats you have reparation but despite that there are certain holes which you have in plug so those holes are the vulnerable points of vulnerability as far as you are concerned and it is a vulnerability that actually hackers exploit so exploit is also a technical word we have discussed this so it is the exploit it is planted through a point of vulnerability and and finally you know that leads to a impact okay so so the impact is also a determinant of risk okay just because there was an attempt on there was actually a attempted attack it doesn't mean that there it had any impact okay so risk also reflects impact okay so keep that in mind so therefore it's a very composite indicator of something that is adward that can potentially happen but what probability exists for it to happen and if it happens what is the impact okay the risk takes into account all these aspect so just keep that in mind risk is not just threat risk is not attack risk is not vulnerability but risk is combination of all this so this is something that you have seen so so you you you have already seen that the formula for risk is actually risk is equal to loss frequency into loss magnitude minus measurement uncertainty okay measurement uncertainty okay in the formula that was given there is also one more factor in the formula which is minus a risk that is already covered by existing measures okay well so well in the new edition of the textbook I checked that they have removed that factor because that is a bit confusing because when you do risk assessment we we have seen in risk assessment there is risk identification risk assessment risk control okay three aspects of risk management so in the risk assessment process essentially you are evaluating only those aspects or we are only evaluating vulnerabilities which exist despite the current controls and therefore it's not very meaningful to have that factor in the formula so if you you can actually go by this formula loss frequency into loss magnitude minus meure uncertainty I think somebody asked this question also in the chat box so and essentially you're always calculating risk as residual risk because the risk assessment process will only capture the risk that currently exist okay after you have set in place certain measures for protection okay despite that there are vulnerabilities that is what you look at and hence you have a valid point in some of one of you asked that question so my response would be this that fact the risk that is calculated using this formula is always the residual R okay I hope that clarifies okay and I think this is one slide which is already covered but this helps you again understand the effectiveness of using biometric devices especially for authentication purposes and they have their own limitations that is something which I stressed in the lecture particularly there are false positives and false negative okay f is false acceptance rate or that is actually false positive how much false positives do occurred okay and F is false negative okay or false rejection ratio and you can see that when false positive is very high false negative is low and Y say vers okay and so therefore there is a optimal meeting point that is where you typically tune the device but despite that you can see that every device will have some false positive and false some false negative in the sense if you use a biometric device you know it has certain sensitivity and it may allow a wrong person and or a the detection can actually be a false positive in the sense you are not the right person but you may be identified as the right person that is false positive false negative is you are the right person but it rejects you so there is chance okay that is what it shows any calibrated devices will have some error okay both on the positive side and on the negative side and the therefore it cannot be the only fullprof for ensuring authentication so therefore one should be aware okay so at what range

one is operating on the devices okay and this is a topic I did not cover in the in the lecture but I thought this will be of interest to many of you because blockchains have emerged as important technology not only in financial transactions cryptocurrencies but also in several aspects of automating organizations using secure Technologies so so the technology actually provides several advantages okay so you must have heard that immutability is an important characteristics of characteristic of blockchain technology and also security okay and a third dimension of this technology is non repudiation okay I thought I will just discuss these three key words in connection with blockchain and how they ensure this we have already discussed what is encryption okay and we have looked at symmetric encryption and asymmetric encryption and how encryption can be used for confidentiality okay so when you generate two keys okay and the public key is used to encrypt a message and when it is transmitted over a channel even if somebody interrupts and gets your message it is encrypted and therefore it cannot be read who can read the an encrypted message the one who has a private key so the recipient has the private key to open or to decrypt the encrypted message and only that person who has the private key can open only therefore a valid recipient has a private key so that is the aspect of confidentiality and that is the sort of security that encryption provides through a transcription a a transaction so if it is a digital transaction and for example you are using your credit card to transact or you are inputting your username and password when it passes through encryption okay so your password or your credit card number or your CBB number your date of birth all are encrypted when it is passed from sender to recipient and therefore your message remain secure okay that is first characteristic of security okay so crypto trans transactions for example provide security thanks to encryption that is point one they also o provide immutability okay so immutability comes through another technology known as hashing okay hashing is basically a mathematical function essentially what hashing means is that when you have a particular content okay and when that content is hashed using a hash function okay it creates an output okay it could be a 256 bit hashed hash output okay so the hashed value would be of course different from the actual content okay but once a Content or the content a transaction data in a blockchain is hashed using a hash function and once the hash is generated now any even a small change in the condent would lead to a totally different hash one and hence if the content is altered okay suppose you made a transaction for say \$100,000 and you try to remove one zero from it okay the hash value changes and therefore it will immediately detect that somebody is trying to manipulate or in other words once a hash value is generated well that is become that becomes a permanent ID for the transaction and you nobody can make a change because that hash value is permanent any new hash value will not be acceptable or it is not the same of course there is security to transactions and the hash value further ensures that your transaction is closed and it is permanent and it is locked with a hash key okay and that is permanent that is the immutability of transactions and the third aspect is non- repudiation which is very important in financial transactions what is non-repudiations actually a legal term okay what do you do in banking transactions when you sign a check for example okay and you want to issue a check to make a payment to a vendor or an individual okay so you have a check issued in your name and what is the ID what are the identifying information in a check of course it has your account number okay and then finally you write to whom you are writing the check and also you write your name and finally you put

your signature actually the main input for authentication that the check is signed by you is your signature okay you know that in a you know in a traditional banking system a check will the signature will be verified with the signature that is stored in the system okay and it is often times manually compared okay in the traditional banking scenario so the signatures are manually compared and if it appears to be the same you know the the one who whom to whom you issued the check receives the money and if the of course signature is different you know then know a flag is raised now suppose you're very clever okay you want to satisfy a vendor that well I have issued a check okay and but at the same time you don't have money in your account or you don't actually really you later decide that you don't want to make the payment or you want to refuse to accept that you have made the signature suppose you made the signature slightly differently and give that check and the accountant or the teller may actually look at your signature and may look may you know well it looks Alik okay but there is some difference okay it's a gray area you later claim in the bank that I did not issue that check somebody else has stolen my checkbook or made the signature it is not me that is called repudiation that is called repudiation you are refusing to accept that you actually made the signature okay Nish I will come to you I know you have a question so so but transactions should be nonre if people keep refusing that they made the transaction then it becomes a real problem in financial transaction so therefore there is a not need for non- repudiation now imagine how private key in in encryption can be used to to ensure non- repudiation okay in a blockchain transaction when you make a transaction your transaction also requires your private key okay or a private key access your digital signature you have already heard this in the lecture okay so the non repudiation is ensured through digital signature or private key okay you sign your transaction with your private key okay or you encrypt your transaction using a private key and hence you cannot refuse that somebody else made that transaction because the transaction accompanies your private key okay so three three characteristics of blockchains are mutability security and non reputation which are actually ensured through Technologies of two technologies encryption and hash and you also know that blockchain is a linked list technically from the data structure point of view so every block actually refers to the previous block through the hash key okay that is generated by the previous block okay so through the keys it is linked okay as a chain so that is a concept of blockchain and each block contains certain standard Fields okay and that Fields pertain to the transaction that is performed and that block as a whole is hash and a key is generated which becomes the identifier for that block permanent identifier that block which is immutable so I thought I will give you an outline of blockchains also because that is a related technology now let me open up for questions I think somebody raised hands can I let me look at chat okay let me first Nish nishar what is your question you have raised hands hello am I audible sir yes a now sir I wanted to ask that we are talking we were talking about blockchain Serv say that again we are talking about blockchain Ser they method of is a method of hashing so those hash functions are Collision resistant functions or normal has Collision resistance okay that's a technical term many people may not be familiar with it that actually ensures the Integrity of and you know the Collision resistance is basically to protect again against hackers so these are all underlying Technologies to ensure security of blockchain transactions okay thank and I would encourage you to credit a course on blockchain which is separately available in NP if you want to learn

more okay now let me look at some questions that are coming up in the chat box somebody asked for sharing the screen okay that is done asks what is the difference between risk and vulnerability okay okay I think this is like you know telling reading the whole Mahabharata and then you finally ask who what's the relation between Rah and Sia but all questions are genuine okay so there are chances that you get confused between dram and CA it is perfectly fine okay so I can tell you that again that suppose okay let me actually use let me actually use this diagram to explain okay so here actually I have tried to show what is what is is vulnerability okay so in the trivial example that I have shown a man is walking on the road with his he has worn a shoe the road is maintained well but unfortunately somebody dropped a banana peel on the road although the road is maintained or protected very well but despite that it happened so happened that there is a banana peel okay so this is the point of vulnerability okay or you can see that this is your asset okay that is what I said so your your computer systems okay your networks okay your you know your the whole infrastructure okay that has a physical wall and then you have you know firewalls and other technologies that protects or reports attacks but despite that there could be vulnerabilities which you haven't fixed okay or for example you haven't updated your Windows systems okay or you haven't updated your M okay so and there may be a new thread because of which the the the manufacturer is actually giving you an update a security update which you actually don't update because there is some downtime required sometimes for updates okay so that that is called a hole in the wall okay a hole in the wall is your vulnerability okay and vulnerability actually paves way for an attack and attacks leads to certain impact certain lws okay and that is the impact so now we have seen we have seen the TVA worksheet you know if you go through the lessons that we learned okay thread vulnerability asset okay sheet okay so risk is calculated for every asset and for every threat so so when you look at a particular asset you find that the asset has certain vulnerabilities like one 2 3 Etc okay it is from that you actually calculate but rest of it is actually well protected you also see that there is a protection against external threads okay but there are also holes okay that is the basis for calculating the loss frequency okay loss frequency is a function of is a function of probability of threat and probability of of success of attack okay success of attack okay success of attack depends on how protected you are okay if you are not vulnerable at all then your success of attack is very very low okay so that is where the connection between the the the risk and the vulnerability comes okay risk is a function of vulnerability but that is not the only variable on which risk depends on okay risk also depends on your impact do you understand that risk is equal to loss frequency into loss magnitude okay so the loss magnitude is derived from the impact who asked this question herani are you clear hello I think somebody asked the question and left it seems so I hope I explained okay so in the calculation of risk vulnerability is only a probability but that is not the only measure that you use to calculate risk okay you also look at the probability of threat and you also look at the impact if that attack happens then you also minus the measurement error from that that is the over old risk that you calate okay so risk and vulnerability are related but not the same okay somebody said we don't have access to the research paper for week one okay which is the paper that you're referring to I think I have given you an online Link we will check this okay out and if we will check if why it is not accessible because always when you access an online resource we put a date to it okay

accessed Dawn you must have seen that okay because an online resource may be accessible today but it may not be accessible tomorrow because the owner can actually withdraw that resource so therefore when we checked in the beginning of this course this was accessible and for some reason if it's not accessible today we will actually try to look for an alternate Source but we cannot guarantee this because these are published papers okay and these papers are actually copyrighted and they also belong to the publisher okay and if somebody is willing to put this up online then we find it okay otherwise we don't so in the NP program we don't actually buy and provide you resources okay if some resources are open and online we try to help you with that otherwise you may have to go to a library to access these resources where it is available. Bal goal can you check this paper for week one and see if you know free link is available and then send it on a chat sure sir or in the chat box okay sir all right okay Nam does that answer question okay I think the one who asked the question is not available okay Niara wants to know what is non repudiation okay repudiation is like this okay you have now typed your name as niara okay and now you sign out okay and then you come in in somebody else's name okay you sign in through somebody else's account okay and suppose that is what and we were keeping track of you and we saw that niharika was here and then she left and then she came back in some other name okay and you come in the name of su and you say I never came in I came in just now okay I didn't ask a question okay that is repudiation you refuse to accept what you did okay now that is not desirable in online banking especially in formal transactions where agents are involved okay X Y these are two agents X and Y are exchanging okay in a financial transaction okay giving and taking but if the giver refuses that I give then there is a problem with the Integrity of the transaction right so you cannot trust the system because people refuse after giving a check that is example I get okay that refusal is technically known as repudiation okay so non repudiation is the quality or the attribute that is required in formal transaction once you transact or once you involve in a formal transaction you should not be able to even if you want to you should not be able to okay able to repudiate okay the system should ensure that there is no fake transaction there is no fake identity okay there is no refusal of transaction and that is very important in financial transaction okay and that is known as non-repudiation yes so does that answer your question yes sir thank you sir okay what are the tools used to identify threats okay that is big question okay you are if you are asking for commercial tools to identify various threats okay so there are multiple types of technology that are actually available to detect and there are technologies that just detect and Report there are also technologies that detect and protect okay malware detection and prevention okay detection is one prevention is one okay if you want to go for both the tools will be more costly okay intrusion detection and prevention okay intrusion detection systems are there that is IDS and there are also intrusion detection and prevention systems okay so there are also firewalls okay we discuss firewalls which are actually Gatekeepers between the inside and outside okay so these are standard technologies that are available for detecting threats and there is also integrative management of threat integrated threat Management Systems today where you know all these tools are integrated into one platform okay so there are different commercial Solutions available unfortunately I'm not an expert in advising you what is the right tool that you should choose Etc please do not embarrass me because that is not you know I don't actually work in that space but if you listen

to the lecture which was given by the external expert you may get some indications about what are some of the tools commercial tools I don't think he also mentions commercial tools because it is not very fair because we don't want to promote any particular company or vendor but we actually talk about the systems in a more generic sense okay okay and nishar you have a question you have raised your hand yes sir I I wanted to ask that in the blockchains we do we use sha algorithms secure hash algorithms which algorithm secure hash algorithm hash algorithm yes that's what I try to explain to you hash is the the heart of command because that is the immutability of blockchains the technology that makes blockchains blockchain transaction immutable is a hash function so once the hash algorithm is applied and a hash value is created the transaction becomes permanent it's like sealing the transaction the hash value is a permanent identifier for a block now you cannot change the contents of the block because the hash value will change so no transaction can be altered once it is made and okay so what is the commonly used Alm like sh or sh2 you're saying something which I'm not hearing sh what do you mean sh 26 26 can you type that in the chat chat box okay sir I did what is the most common oh don't oh okay the so I'm not I can't tell you this because you're asking about what is the most commonly used algorithm sha 256 I think this has to do with the number of bits right could be 256 bit it could be 512 Etc I I assume so but this I should not answer this question because I may not be I'm not very updated on this but well hash algorithms are used I encourage you to take a course on blockchain separately my effort was only to link cyber security and the blockchain as a specific case of it okay hash functions is not something that I cover here that is a lot of mathematical foundations okay and you have to build on build from that then you can actually look at what are the current algorithms that are popular in use Etc okay so I can't deviate to that I'm not prepared also sorry about it okay sir thank okay and nishar you so that was your question go hands down now yes okay I think Bal goal has responded to nam's question okay all right okay and also RI has a question can we have case studies also we go through the presentation lecture but going through the case studies will help yeah again actually you're coming back to the same topic which actually I said in the first lecture itself the original cases like the Howard business school cases or research papers Etc have commercial value and copyrights okay so formally we should not and we cannot share the original papers with you on any platform but that is why we actually try to give you links to the case studies even Howard case studies some of them are available online you know for example the case we googled you which is coming up it's available online it's free so those cases you can access and some Snippets of this case studies are available in other sites which we searched out and in the course outline you have received links to that apart from that we are helpless so forgive and the other ways if you your institution has subscription to Howard case or you know your friends institution has Etc when I was a research scholar that is how I used to get research papers okay my institution did not have subscription to all the papers that I want okay often times you reach this dilemma well I search for it it's not subscribed by my digital library no even in in I Madras I face that problem okay informs you know which is an important set of journals we don't have access to it okay so what do I do I ask my friends can you actually do that so I get it from Sister institutions okay don't don't be ashamed in asking for resources from friends and you know colleagues where they may have access to okay that is how we learn so please please search for Alternatives and we can't

provide everything to you and that has a cost and that also has legal implications okay Rajiv okay and you may also have noticed wherever there is a commercial case involved and we have not given you a copy we do not ask questions on that case in the assignments okay to be fair to everyone but that doesn't prevent you from learning the Context of cases okay and namira ask sir could you please suggest some sources to refer pyqs I didn't get what is a pyq what do you mean by pyq the are you there you set pyq I previous year question previous question oh okay I know FAQ but PQ I'm sorry I was not so familiar with it previous year question papers well this is not a resource I can provide because this will depend on the policy of the NL platform you can write to them what they do about previous year questions because this is a fairly new course okay this is a second run so therefore you know the the the question bank is limited and you can see that you know as far as I know there is no course on information privacy in N okay it's a new phenomenon it's a new course and it's very difficult to to give you questions because they are not in plenty okay and not many years have passed after the course started it's a new course okay so bear with us and go through the lessons go through the Powerpoints go through the resources that you can access and you will be fine as far as your exams are concerned okay somebody askings provide me with names of Open Source sites for most recent research articles and journals on cyber security well this is again interesting you know all of you are aware of scholar.google.com okay so we call it Google Scholar Google Scholar does this job of aggregation okay if you if you search for a paper in Google scholar not in Google search but in Google Scholar if you search for a paper that is actually a very targeted search okay and they do this aggregation right if that paper is posted by anyone anywhere in any site Google Scholar will tell you okay and you can it will give the link and you can actually download the paper from that site okay and other other source is research gate okay okay let me actually get the actual address give me a moment r okay I have shared the site address here researchgate.net I believe some of you are already members in research gate and if you're are not a member I encourage you to become a member or you sign in with your credentials to research gate and in research gate you can actually search for research papers and sometimes the paper is readily available the authors have shared it here it's an author Network basically it's a scholar Network okay so and sometimes the paper is not available then you then you can actually search for the names of the others in research gate and you can actually send a request to the order to share the paper and you know generally an author will share one's paper although it U doesn't so the interest of the publisher but an author may be willing to share a version of that paper sometimes which was a version before it was published because there may be copyright issues so they will usually share that paper if somebody asked me a for a paper I share it there is a incentive for sharing that is Citation okay you may actually read my paper and cite me in your paper in your work so so keep that in mind research gate is another place to go to get research papers I hope that answers your question so I've shared two sites call. google.com and research g.net okay and there is a last question I think we are almost done how the Privacy techniques can be utilized in real world scenario py mentioned some scenario especially for banking sector okay so I think you are jumping the gun now so you have the privacy module starting now okay from next week onwards it is actually on information privacy so and information security and information privacy are closely related and you will learn about information privacy in the digital world

from next week onwards okay so you can defer this question for the next live interaction okay which which will be in the next month and if you still have questions on privacy you please ask that time so you will definitely go through foundations of information privacy and you may actually get satisfactory answers very soon that's what I believe and let me also look at the questions that were shared by you in the chat box I think we already answered most of the questions problem okay I'm sorry okay there was a question in the case of ransomware attack how will the company pay the demanded amount to attack generally for my limited knowledge they may pay to their bank account net banking credit debit UPI wallet I heard about the cryptocurrency about which I have no knowledge if the pays in methods mentioned in the option one cyber crime police can easily Trace them and arrest them right as cyber security student I'm curious thank you well good question as to how you make payment to a hacker because in ransomware Attack the hacker has taken control of your machine machine is encrypted and the key is with the hacker okay and you need that key to decrypt the machine you know somebody else logged your house and you know standing with a key okay you pay money and then the key will be given to you now this is a problem so on one side your house is locked or your business is down you can run your you cannot run your systems or your e-commerce is down okay and you are actually losing revenues and it is an illegal attack okay so it's a thief who is at the door and who is actually blocking your resources and hence you can of course you know look for remedies but today when you learn again on policies you know particularly privacy policy and all you will find that if there is a hacker attack or if there is a cyber security incident it has to be reported to a competent Authority okay in today in India there is something called digital personal data protection act or dpdp which again requires a an attack to be reported in a stipulated period of time in gdpr it is 74 hours within 74 hours an incident should be report and hence what that means is that if there is a breach or if there is a ransomware attack you are supposed to report it first okay report the incident first okay and create a report proper report and report it to the police report it to the competent Authority and then you are safe because you have to restore your operations okay and you are paying a thief and if somebody is placing a barrel on your head and ask for money you know you have to protect your life okay you may actually give money and protect yourself okay that is a kind of condition okay so I think the U law will not punish you because you are trying to protect yourself okay I'm not a expert of the law here but as far as I understand to the best of my knowledge paying a hacker to save your business after reporting okay you must be Sav but I I'm not here speaking as a legal expert but keep in mind that you should actually create enough evidence of a Cyber attack and report it and then protect yourself I hope that will be my answer that that that satisfies you but this is also an accounting problem and experts in that area should actually answer you saying what is the best mod of payment and often times you may not have the option because the hiker would say pay by cryptocurrency alone okay then you have to pay in that otherwise you don't get the key to open your system right so you succ and I think the question number two on calculating residual risk I explained in the beginning itself and that is answered already all right and is there any other question okay there are two questions I want to know about the opportunities for a resarch SC Industries in India let me again tell you this is something related to carrier right and it's not fair for me to actually give any advice on Career because that is a separate specialization and it requires a

lot more inputs from an individual before any advice can be given okay so you may be misguided if you say oh lot of opportunities inia go to Amazon go to Google you know but it also depends on you know your specialization it's always a match okay between your competency and what the other side is looking for yeah so I would say research Scholars do have opportunities in research for in in the research wings of organizations but check out check out what kind of qualifications and experience are they looking for okay so you know from I would say in in Indian context you know in academic research okay we have research programs most of our research scholar for their PHD they go to an academic institution but some some of them of course go to Industry and they find jobs in you know specialized areas where research is active in that organization Sanjay has asked one person asked a Iran Commander was killed by a us drawn how is it possible how do we answer this question the drawn actually killed okay and you can read about it in you know in newspapers okay in public domain I just reported what I read in the public domain and it is a US draw that kill and I think human oids you must be reading a more a little more on current technology to what is going on today you can actually send a humanoid to an enemy paret could be a be or could be an insect that is going to the other side but it will be carrying very you know you know exploits so a lot of new developments are happening in technology drawn is one of them and it is possible to use drawn against individual okay it's a weapon in a war today rega this may be yeah we are turning one: so I will close it one last question is there some document available on how draft policy on data protection and privacy compare with gdpr well well there is a discussion coming up on both okay our India's DP DP dpdp so they call it digital personal data protection act which has become which became an act last year and also gdpr okay so you will see student groups actually discussing this in the upcoming sessions okay you can wait for that so we do have sessions on both and you will be able to get some comparison of the two so you have to wait for the next module I think I have covered all your questions in the chat box in Zoom ASAS the questions that came in the track that was opened for this live interaction in Google okay and if you still have questions feel free to ask us all your questions in the chat box and we will be responding to you and we look forward to receiving more feedback more questions and more things more responses from you about the course okay enjoy this course and all the best and I'm looking forward to meeting you all in the next live interaction and reserve your questions some of reserve some questions for the next live interaction thank you very much to all of you and have a good day thank you sir thank you thank you very much sir thank you sir e e e e e e e e e e e e e e e e for